Personal Budget Tracker
Database Design Document

CP476 - Internet Computing

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February 2026

https://www.figma.com/make/07E7gZdfvxFnv6w1t1vA4d/Create-ER-Diagram?t=o5b0GWKSVOMlmNYg-1

SQL CREATE TABLE Statements

```
CREATE TABLE users (
user_id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(120) NOT NULL,
email VARCHAR(255) NOT NULL UNIQUE,
password_hash VARCHAR(255) NULL,
created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

```
CREATE TABLE categories (
category_id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(80) NOT NULL,
icon VARCHAR(120) NULL,

user_id INT NULL,

created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT fk_categories_user
FOREIGN KEY (user_id) REFERENCES users(user_id)
ON DELETE CASCADE,

CONSTRAINT uq_category_name_per_user
UNIQUE (user_id, name)
);
```

```
CREATE TABLE expenses (
expense_id INT AUTO_INCREMENT PRIMARY KEY,

user_id INT NOT NULL,
category_id INT NOT NULL,

amount DECIMAL(10,2) NOT NULL,
expense_date DATE NOT NULL,
description VARCHAR(255) NULL,
payment_method VARCHAR(60) NULL,

created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT fk_expenses_user
FOREIGN KEY (user_id) REFERENCES users(user_id)
ON DELETE CASCADE,

CONSTRAINT fk_expenses_category
FOREIGN KEY (category_id) REFERENCES categories(category_id)
```

```
ON DELETE RESTRICT,

CONSTRAINT chk_expense_amount_positive
CHECK (amount > 0)
);
```

```
CREATE TABLE budgets (
budget_id INT AUTO_INCREMENT PRIMARY KEY,

user_id INT NOT NULL,
category_id INT NULL,

limit_amount DECIMAL(10,2) NOT NULL,

budget_month DATE NOT NULL,

created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT fk_budgets_user
FOREIGN KEY (user_id) REFERENCES users(user_id)
ON DELETE CASCADE,

CONSTRAINT fk_budgets_category
FOREIGN KEY (category_id) REFERENCES categories(category_id)
ON DELETE SET NULL,

CONSTRAINT chk_budget_limit_positive
CHECK (limit_amount > 0),

CONSTRAINT uq_budget_user_month_category
UNIQUE (user_id, budget_month, category_id)
);
```